Model Evaluation

1. Introduction

Model evaluation is performed to measure how well the trained machine learning model predicts loan default risk. In this project, a Random Forest Classifier was used to predict whether a borrower will default on a loan. The model was evaluated using multiple performance metrics such as accuracy, confusion matrix, classification report, ROC curve, AUC score, and Gini coefficient.

The dataset was divided into training and testing sets using an **80–20 split**. The model was trained on the training data and evaluated on the testing data.

2. Accuracy Score

Accuracy measures the proportion of correctly predicted observations to the total observations.

Formula

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Model Accuracy

```
Accuracy = 0.74
```

This indicates that the model correctly predicts approximately **74% of the loan default cases**.

3. Confusion Matrix

The confusion matrix provides a detailed breakdown of correct and incorrect predictions.

Output

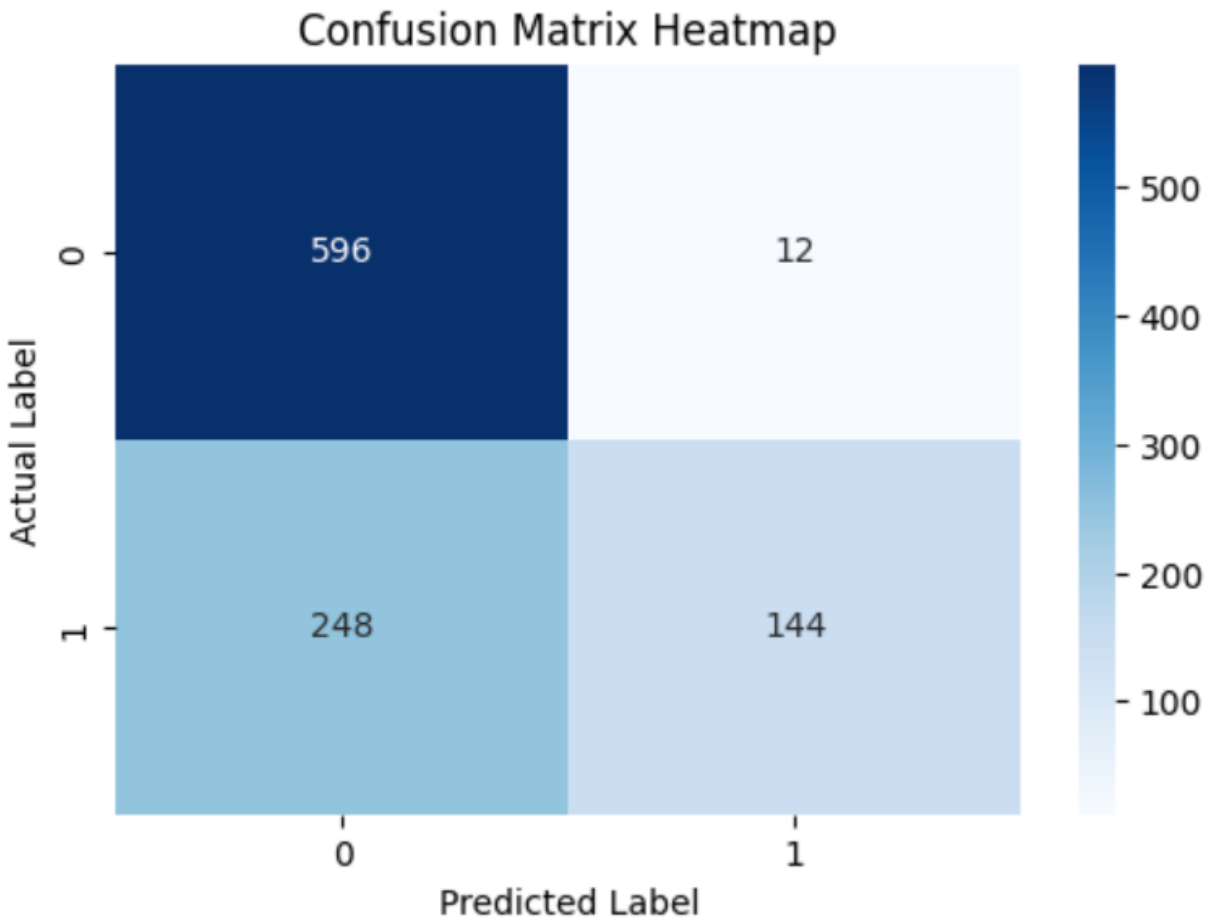
```
[[596  12]
 [248 144]]
```

Where:

Value	Meaning
596	True Negatives (correctly predicted non-default)
144	True Positives (correctly predicted default)

Value	Meaning
12	False Positives
248	False Negatives

Confusion Matrix Visualization:



This visualization helps understand how many borrowers were correctly or incorrectly classified by the model.

4. Classification Report

The classification report provides precision, recall, and F1-score values.

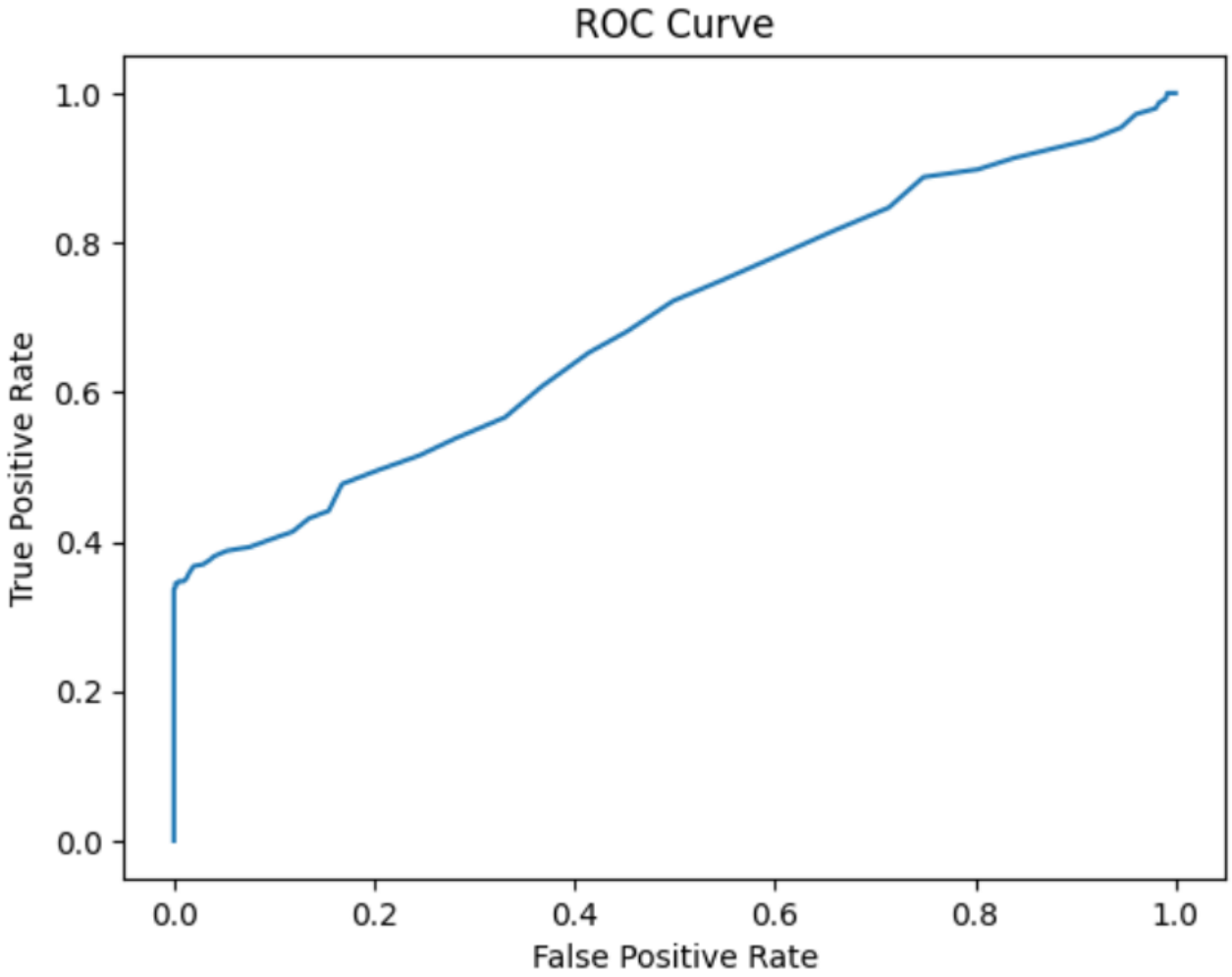
- **Precision** measures the accuracy of positive predictions.
- **Recall** measures the ability of the model to identify actual default cases.
- **F1 Score** is the harmonic mean of precision and recall.

These metrics help evaluate how well the model identify high-risk borrowers.

5. ROC Curve

The Receiver Operating Characteristic (ROC) curve illustrates the diagnostic ability of the classifier by plotting:

- True Positive Rate (TPR)
- False Positive Rate (FPR)



6. AUC Score

The Area Under the Curve (AUC) represents the probability that the model correctly distinguishes between defaulters and non-defaulters.

AUC Score = 0.693

An AUC value close to 0.7 indicates that the model has moderate predictive ability.

7. Gini Coefficient

The Gini coefficient is widely used in credit risk modeling.

Formula: $\text{Gini} = 2 \times \text{AUC} - 1$

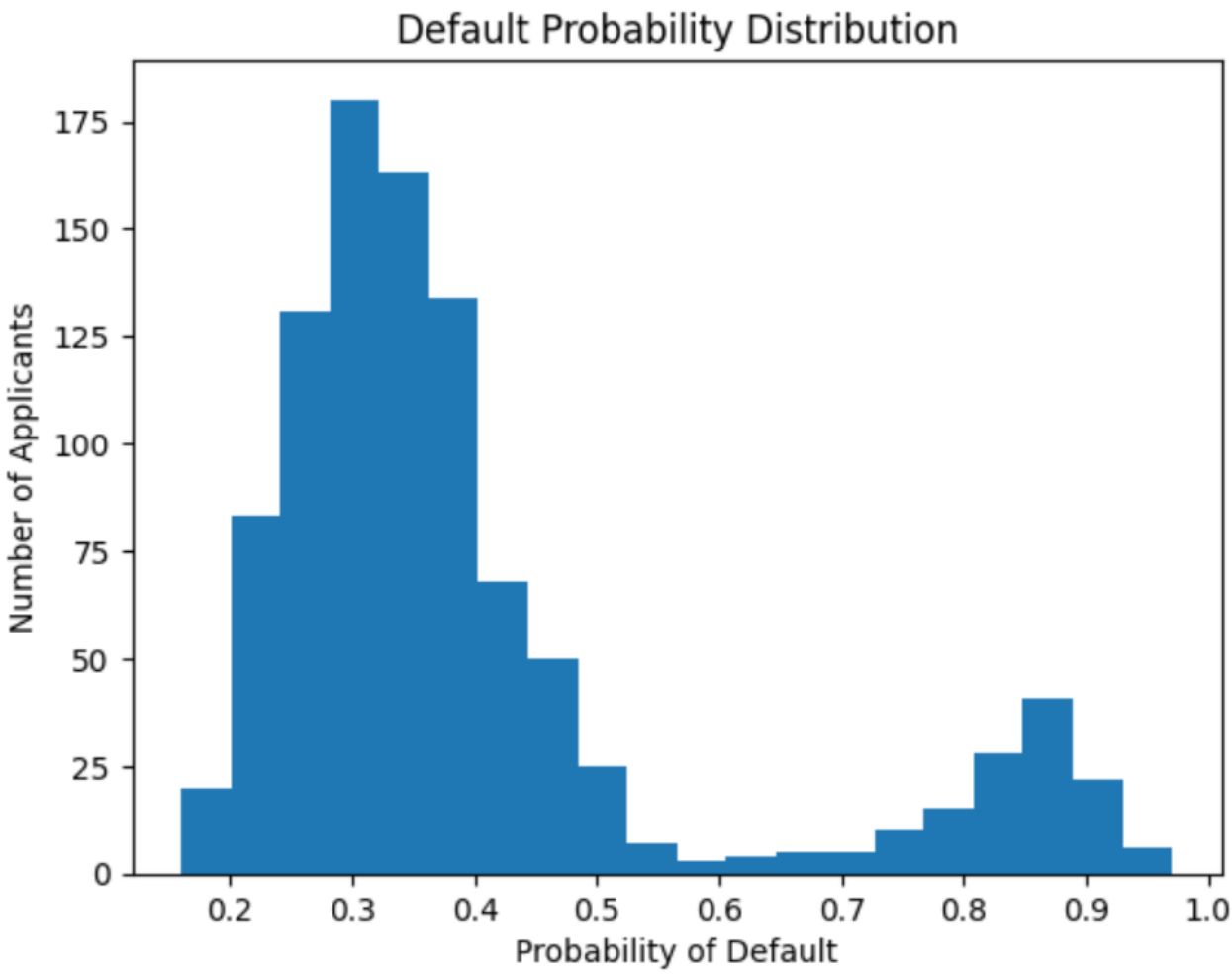
Calculated Gini Value:

Gini = 0.387

A higher Gini value indicates stronger predictive performance.

8. Probability Distribution of Default

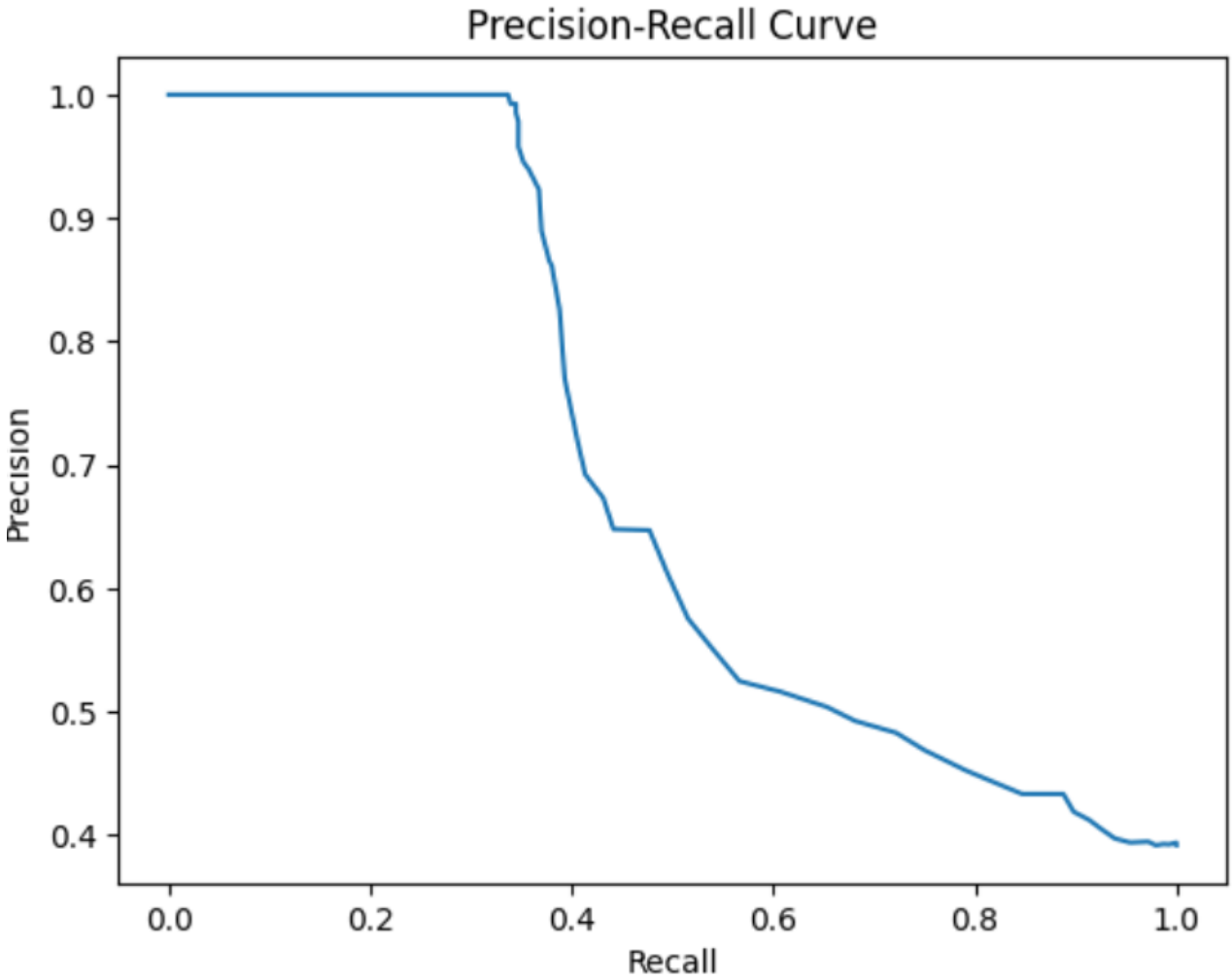
The predicted probability distribution shows how the model assigns risk scores to borrowers.



Borrowers with higher probability values are considered high-risk applicants.

9. Precision–Recall Curve

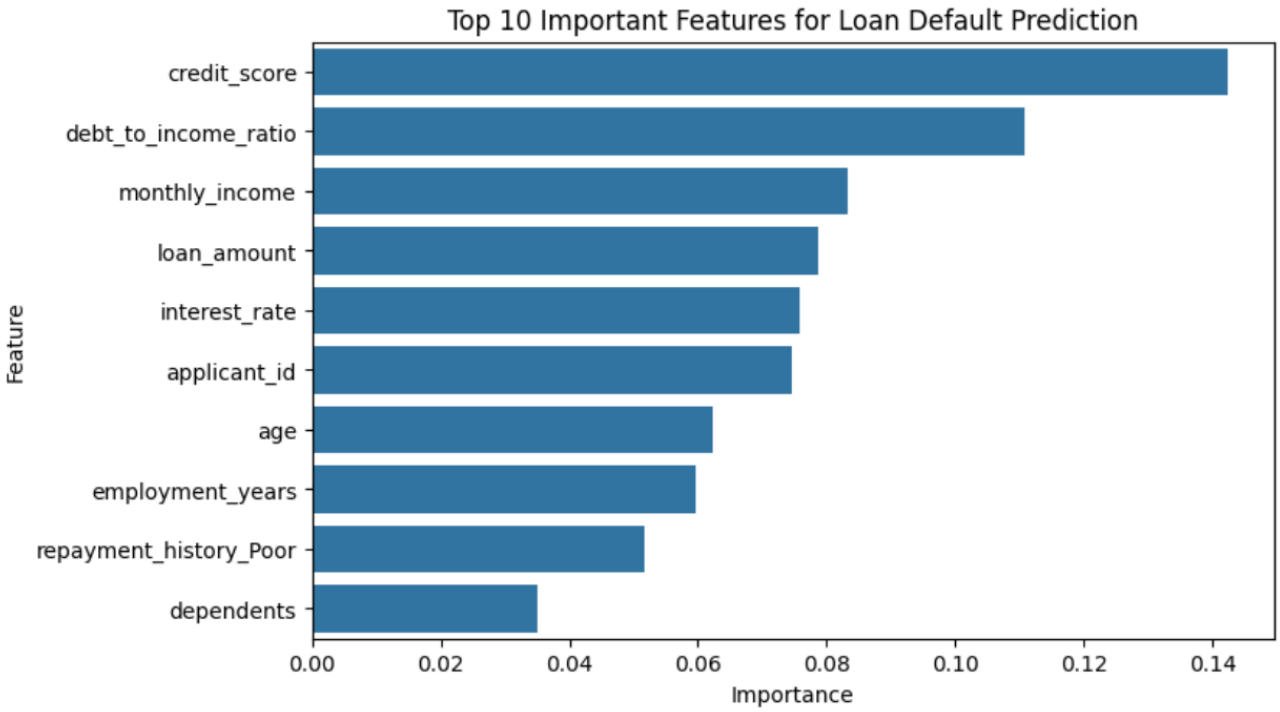
The precision–recall curve evaluates the model's ability to detect positive cases (loan defaults).



This curve is especially useful for imbalanced datasets, where the number of non-default cases is larger than default cases.

10. Feature Importance Analysis

Feature importance analysis helps identify the variables that have the greatest influence on the model's predictions. Using the Random Forest model, the importance of each feature was calculated and the top contributing factors were visualized.



The graph shows the ten most influential features affecting loan default prediction. These features provide insight into which borrower characteristics play a major role in determining loan repayment risk, helping financial institutions understand key risk indicators.

11. Relation to SDG Goal – No Poverty

This project contributes to Sustainable Development Goal 1 – No Poverty by supporting better decision-making in microfinance lending.

- The machine learning model predicts whether a loan applicant is at high or low risk of default, helping financial institutions assess borrowers more effectively.
- By identifying risk accurately and adjusting the classification threshold, the system helps promote fair lending practices and improves access to financial services for individuals in developing regions, thereby supporting poverty reduction and financial inclusion.

12. Conclusion

The Random Forest model achieved an **accuracy of 74%**, demonstrating a reasonable ability to classify loan default risk. The confusion matrix and ROC analysis show that the model can distinguish between default and non-default borrowers with moderate performance. These results suggest that the model can assist financial institutions in identifying high-risk borrowers and improving lending decisions.